



University
of Windsor

MASTER OF ENGINEERING

Computer Engineering

Ontario, Canada

Graduate coursework, projects and research

14

REPOSITORIES

4

SEMESTERS

2

MAJOR PROJECTS



University of Windsor
Faculty of Engineering · Ontario, Canada

Computer Engineering



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DEGREE
Master of Engineering
SEMESTERS
I – IV
REPOSITORIES
15 indexed
SESSION
2023 – 2024

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A Note from Amey Thakur

Education is not a destination but a continuous journey. This repository represents my commitment to that philosophy: a deliberate effort to preserve, organize, and revisit the advanced knowledge acquired during my graduate studies in Computer Engineering.

The creation of this archive stems from a fundamental belief: **true learning transcends the classroom and extends far beyond graduation**. As I navigate my professional career, I recognize that the advanced concepts, methodologies, and problem-solving frameworks developed during these formative years remain invaluable. However, knowledge, when not actively maintained, fades. This repository serves as my intellectual foundation: a structured collection I can return to for relearning, reference, and reflection.

Why this repository exists

Knowledge Preservation	To capture and maintain the depth of understanding developed across four semesters of rigorous graduate study.
Continuous Learning	To create a resource that supports lifelong learning, enabling me to revisit advanced principles as I encounter complex challenges.
Academic Integrity	To document my authentic academic journey - every concept studied, every project implemented, every problem solved.
Open Contribution	To share these materials with the broader community, believing that knowledge grows when freely exchanged.

This is more than a collection of files or a digital archive. It is a testament to years of intellectual growth, a record of challenges overcome, and a foundation upon which future learning will be built. By making this work publicly available, I hope it serves not only my own continued education but also assists others on their learning journeys.

Note

All materials in this repository were created, compiled, and organized by me throughout my graduate program (2023-2024) as part of my coursework, laboratory assignments, and project implementations.

— Amey Thakur

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Start Here

What is in here.

14 REPOSITORIES	2 MAJOR PROJECTS	4 SEMESTERS	0 SUPPORT RESOURCES
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Who it is for

You are on the same syllabus

Every subject is here in the order it was taught, with the notes, the laboratory work and the reports that went with it.

You are teaching yourself

The projects come with working code and, for many of them, a recording of the thing running.

You are assessing my work

This is the whole record, linked and dated, with nothing curated out.

Tip

Everything linked from this document is published under Creative Commons Attribution 4.0. You may read it, copy it, adapt it and build on it, including commercially. Credit me and you are done.

Four ways in

Browse the coursework

Every subject I took, in semester order, each linked to its repository.

[Semester sections](#)

Go straight to the projects

The work that went past a weekly assignment, with what each one does.

[Major Projects](#)

Watch it running

Recorded demonstrations of the projects and laboratory sessions.

youtube.com/@ameythakur

Find me elsewhere

Publications, code, profiles and everything outside this archive.

amey-thakur.github.io

How to Use This Guide

It is an index, not a textbook	Every row names a repository and links to it. The repository holds the material.
Semester order, not alphabetical	Subjects appear as I took them. A later course usually assumes an earlier one.
Course and laboratory share a repository	Where a subject had both, both codes point at the same repository.
Projects are listed on their own	Substantial work has its own section, because it did not take the same effort as a weekly assignment.
The repository is the live copy	This document is dated. Where the two disagree, the repository is right.

! Important

This is my own archive. It is not a course, a curriculum, an accredited programme, or an official publication of the University of Mumbai, Terna Engineering College, or the University of Windsor. The institutional marks appear only to place the work in context.

M.Eng. Computer Engineering

Four semesters at the University of Windsor: core engineering subjects, with electives in machine learning, computer networks, computational intelligence and digital communications.

Degree	M.Eng. Computer Engineering
University	University of Windsor
Structure	Four semesters
Repositories indexed	14
Semesters with archived work	4

Semester I

Code	Subject	Repository
GENG 8000	Engineering Technical Communications	ENGINEERING-TECHNICAL-COMMUNICATIONS Course Assignments · Weekly material, 10 weeks · Reference Books · Discussion Posts · Office Hours
GENG 8010	Engineering Mathematics	ENGINEERING-MATHEMATICS Course Assignments · Weekly material, 11 weeks · Final Exam · Midterm Exams · In-Class Quizzes · Reference Books · WileyPLUS

Semester II

Code	Subject	Repository
GENG 8020	Engineering Project Management	ENGINEERING-PROJECT-MANAGEMENT Course Assignments · Notes · Reference Books · Lecture Slides
GENG 8030	Computational Methods and Modeling	COMPUTATIONAL-METHODS-AND-MODELING-FOR-ENGINEERING-APPLICATIONS Course Matlab Project · Announcements · MATLAB Tutorials and Solutions

Project work

[ADAPTIVE-CRUISE-CONTROL](#)

Adaptive Cruise Control (ACC) with Arduino, MATLAB, and Simulink · Real-time speed regulation and proximity-based safety system

Semester III

Code	Subject	Repository
ELEC 8560	Computer Networks	COMPUTER-NETWORKS Course Assignments · Homework · Lecture Notes · Reference Books
ELEC 8900	Machine Learning	MACHINE--LEARNING Course Assignments · ML Project · Weekly material, 9 weeks · In-Class Presentation · Reference Books · Course Project Instructions · datacamp
VIP-CSL	VIP - Community Service Learning	VIP-CSL-FALL-2023 Course Content Modules · Manual & Contract · Submissions · Timeline

Project work

[ZERO-SHOT-VIDEO-GENERATION](#)

Zero-Shot Video Generation, A neural video synthesis studio powered by latent diffusion and cross-frame attention

Semester IV

Code	Subject	Repository
ELEC 8330	Computational Intelligence	COMPUTATIONAL-INTELLIGENCE Course Assignments · Final Exam · Reference Books · EAB · MATLAB · Midterm Exam
ELEC 8900	Digital Communications	DIGITAL-COMMUNICATIONS Course Homework · Lecture Notes · Notes · Reference Books · Project · Tutorials

Major Projects

Work that went past a weekly assignment. Each entry gives the subject and semester it came from.

Zero-Shot Video Generation

Machine Learning & Computer Vision (Semester III)

Zero-Shot Video Generation, A neural video synthesis studio powered by latent diffusion and cross-frame attention

[ZERO-SHOT-VIDEO-GENERATION](#)

Adaptive Cruise Control

Computational Modeling & Control Systems (Semester II)

Adaptive Cruise Control (ACC) with Arduino, MATLAB, and Simulink · Real-time speed regulation and proximity-based safety system

[ADAPTIVE-CRUISE-CONTROL](#)

Student support resources

Services I used as an international student. Part of the degree in practice, and hard to find otherwise.

International Student Centre

International Student Centre (ISC), University of Windsor · Orientation Resources & Student Support Hub

Engineering International Student Advising

Engineering International Student Advising (ENG-ISA), University of Windsor · Academic, immigration, and student success resources for international engineering graduate students

Writing Support

Writing Support Desk (WSD), University of Windsor · Academic writing, citation standards (APA, MLA, IEEE, Chicago), and academic integrity resources

Project Demonstrations

Recordings from my own channel, made while the projects were still running.

Tip

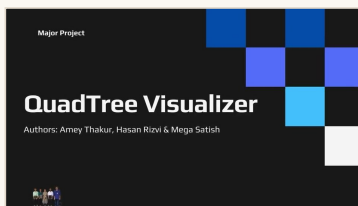
Every thumbnail below is a link. 17 videos, 1 hour 25 minutes in total, on my own channel.

Full playlist

youtube.com/playlist?list=PLG0c13Pt035Z9INe4gyxoZnA4zAXssiLn

Channel

youtube.com/@ameythakur



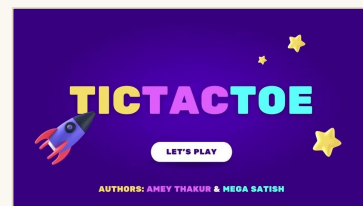
QuadTree Visualizer

4:58



Zero-Shot Text to Video Generation

3:57



Tic-Tac-Toe Game in Angular Framework

4:06



Calculator App using Android Studio

4:59



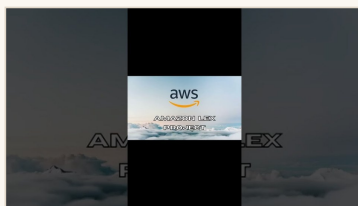
Flashlight App using Android Studio

2:04



Online Chess Game

1:27



Pizza Ordering Chatbot using Amazon Lex - AWS Project Demo

1:53



Pizza Ordering Chatbot using Amazon Lex - Presentation

1:11



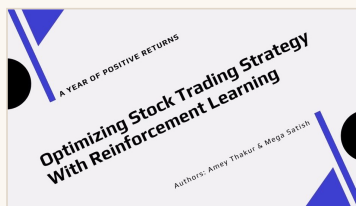
Pizza Ordering Chatbot using Amazon Lex - AWS Project

42:15



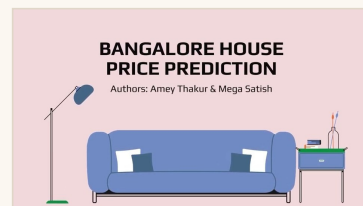
Text Summarizer

1:08



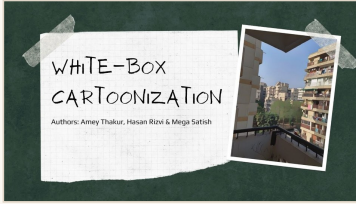
Optimizing Stock Trading Strategy with Reinforcement Learning

1:24



Bangalore House Price Prediction

1:33



White-Box Cartoonization

3:03



DeepFake Audio

3:30



Digital Bookstore

4:40



Chat Room using HTML, PHP, CSS, JavaScript, AJAX

2:29



Hangman Word Game using Applet in Java

1:14

Repository Directory

Every repository in this document, A to Z. Each name is a link.

Repository	Subject or project	Where it sits
ADAPTIVE-CRUISE-CONTROL	Adaptive Cruise Control (ACC) with Arduino, MATLAB, and Simulink · Real-time speed regulation and proximity-based safety system	Semester II · Project
COMPUTATIONAL-INTELLIGENCE	Computational Intelligence	Semester IV · ELEC 8330
COMPUTATIONAL-METHODS-AND-MODELING-FOR-ENGINEERING-APPLICATIONS	Computational Methods and Modeling	Semester II · GENG 8030
COMPUTER-NETWORKS	Computer Networks	Semester III · ELEC 8560
DIGITAL-COMMUNICATIONS	Digital Communications	Semester IV · ELEC 8900
ENGINEERING-INTERNATIONAL-STUDENT-ADVISING	Engineering International Student Advising (ENG-ISA), University of Windsor · Academic, immigration, and student success resources for international engineering graduate students	Student resource
ENGINEERING-MATHEMATICS	Engineering Mathematics	Semester I · GENG 8010
ENGINEERING-PROJECT-MANAGEMENT	Engineering Project Management	Semester II · GENG 8020
ENGINEERING-TECHNICAL-COMMUNICATIONS	Engineering Technical Communications	Semester I · GENG 8000
INTERNATIONAL-STUDENT-CENTRE	International Student Centre (ISC), University of Windsor · Orientation Resources & Student Support Hub	Student resource
MACHINE--LEARNING	Machine Learning	Semester III · ELEC 8900
VIP-CSL-FALL-2023	VIP - Community Service Learning	Semester III · VIP-CSL
WRITING-SUPPORT	Writing Support Desk (WSD), University of Windsor · Academic writing, citation standards (APA, MLA, IEEE, Chicago), and academic integrity resources	Student resource

ZERO-SHOT-VIDEO-GENERATION

Zero-Shot Video Generation, A neural video synthesis studio powered by latent diffusion and cross-frame attention

Semester III · Project

What I Would Tell You

Six things I learned building this, offered for whatever they are worth.

1 Keep one repository per subject.

Not one per assignment and not one for the whole degree. A subject is the unit you will actually search for later.

2 Write the report in the same week.

I wrote every laboratory report while the work was still in front of me. Reconstructing one a month later takes longer and it comes out thinner.

3 Record it while it still runs.

Environments rot. Several of my demonstrations exist only because I recorded the session on the day, and the code that produced them will not build now.

4 Take the notes you would want in five years.

Not the ones that get you through the examination. Those two sets of notes look different, and only one of them is worth keeping.

5 Publish it.

A private archive rots quietly. Making mine public is what forced it into a state someone else could follow, and that discipline is most of the value.

6 Index the whole thing.

Material nobody can navigate is a folder, not an archive. The index is the part that turns four years of files into something usable.

💡 Tip

If you take one thing from this: the work is only worth as much as your ability to find it again.

Explore Further

Where the rest of the work lives.

Related repositories

COMPUTER-ENGINEERING	The B.E. index this document is drawn from
MENG-COMPUTER-ENGINEERING	The M.Eng. index this document is drawn from
ACHIEVEMENTS	Certifications, course completions and awards
RESEARCHGATE	Code and artefacts accompanying published work
YouTube playlist	Engineering Projects, video demonstrations

Elsewhere

GitHub	github.com/Amey-Thakur
Amey's Arc	amey-thakur.github.io
ORCID	0000-0001-5644-1575
Google Scholar	Publications and citations
ResearchGate	researchgate.net/profile/Amey-Thakur
arXiv	arxiv.org/a/thakur_a_3
Kaggle	kaggle.com/ameythakur20
Hugging Face	huggingface.co/ameythakur
LinkedIn	linkedin.com/in/amey-thakur
YouTube	Engineering project demonstrations

A Closing Note

The repositories are the work. This is the index to them, written so someone arriving without context can find the part they need.

! **Important**

Repositories change. Where this document and a repository disagree, the repository is right.

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